Modeling and evaluation of thermionic energy converters in the space-charge limit

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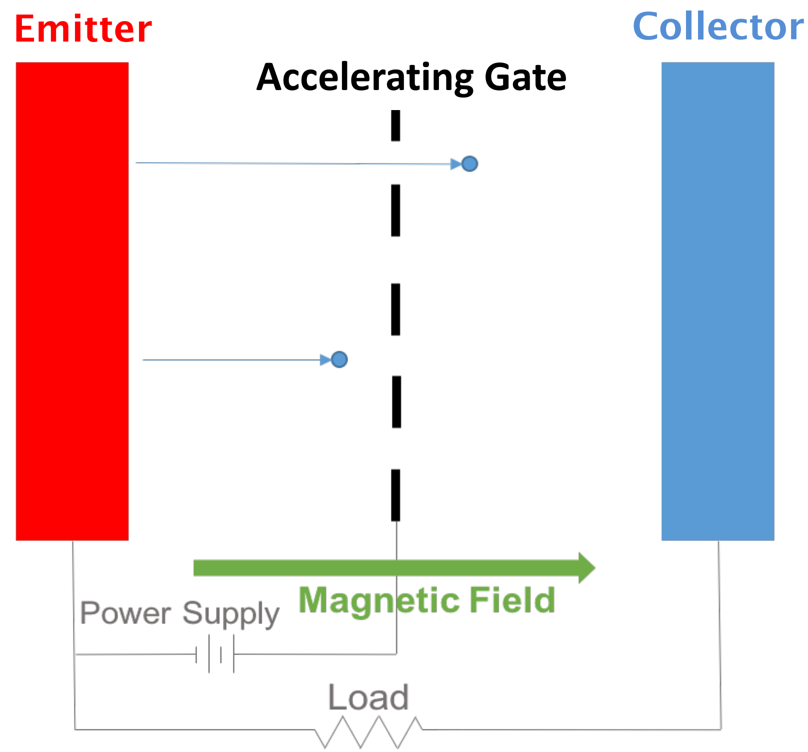
September 3, 2019

2019 North American Particle Accelerator Conference

This work is supported the US DOE Office of Science, Office of High Energy Physics: DE-SC0017162.

Thermionic energy converters

- Thermionic Energy Converters transform heat to electricity
  - *Thermionic emission drives current across an external load*
    - High theoretical efficiency
    - Scalable (nm – mm gaps)
    - Tile-able (m scale surface area)
    - Tunable (1200-2000° K)
    - Robust



- Simulation fidelity lags materials development and testing
  - *High computational expensive!*
    - Electron surface interactions require millions of macro-particles
    - Strong model dependence with tunable parameters
  - *Large measurement discrepancies!*
    - Measurements should guide modeling process

Improvements in modeling TECs using the Warp Code

- An open-source* plasma and accelerator simulation framework, developed by Lawrence Berkeley National Laboratory

- Flexible 2D, R-Z, and 3D electrostatic and electromagnetic particle-in-cell
  - *Emission: multi-species particle emission with different physics models*
  - *Internal conductors, dielectrics, applied external magnetic fields*
  - *Scalable to 10,000+ core operation*

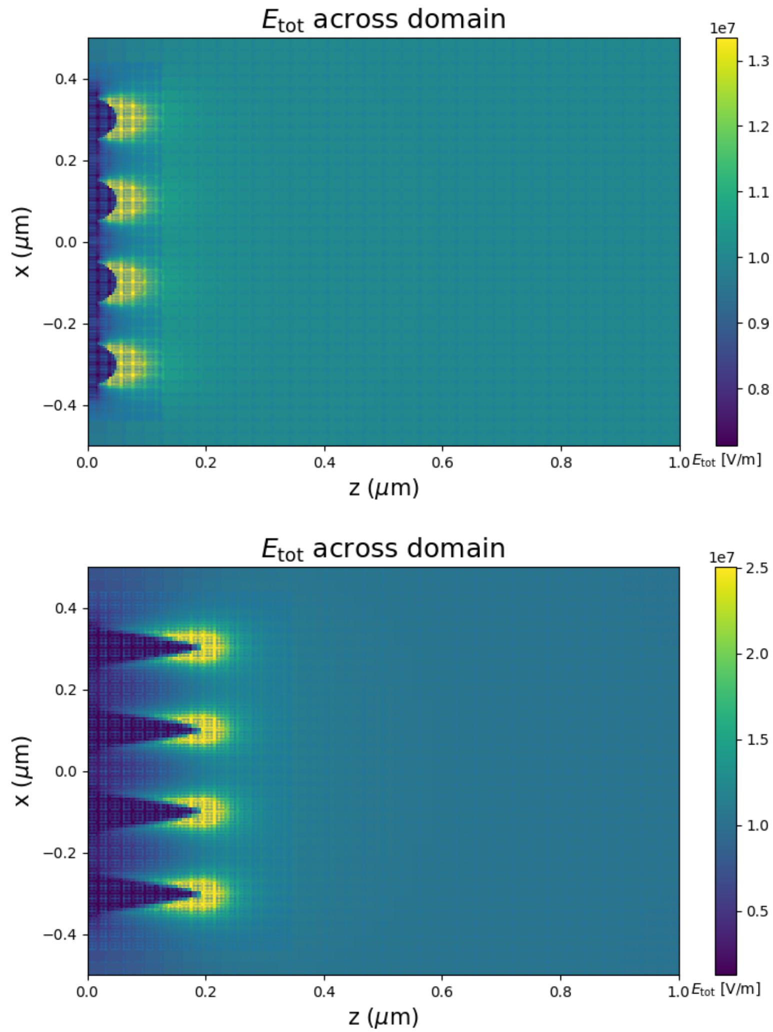
- RadiaSoft efforts to support vacuum devices:
  - *CAD input-output with support for complex geometries via STL*
  - *Improved emission models for novel cathodes*
  - *New geometry capabilities (mesh-refinement/"cut-cells" at internal boundaries)*
  - *Particle Reflection Models*

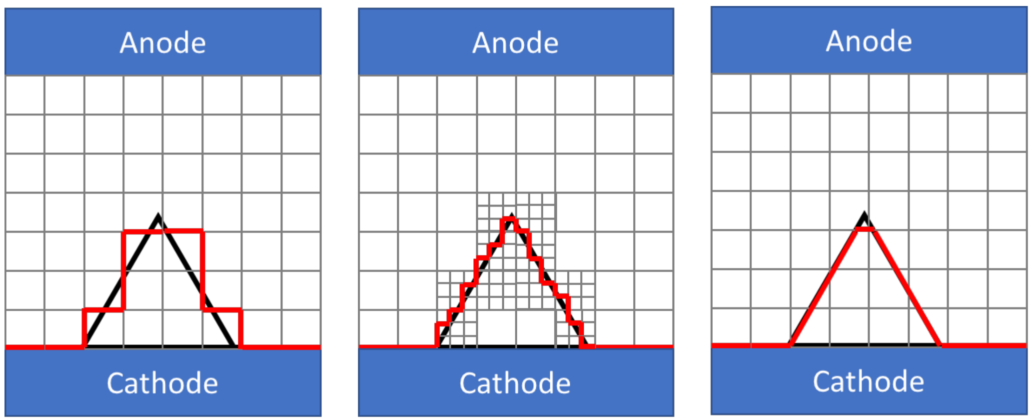
- Source code available at the following locations:
  - *Berkeley Lab Warp: https://bitbucket.org/berkeleylab/warp/src/master/*
  - *RadiaSoft Fork: https://bitbucket.org/radiasoft/warp/src/master/*
  - *RadiaSoft Python Tools for Warp: https://github.com/radiasoft/rswarp*

Modeling emission in Warp

- TECs are macroscopic thermionic emission devices.
  - *Particle statistics are required for fidelity*
- Adapt PIC to be more accurate
  - *Mesh refinement for field sampling*
  - *`Cut' cells for sub-cell boundaries*

$$j = A_G T^2 e^{-\phi_{\mathrm{e}}/k_{\mathrm{B}}T} e^{-\left(\sqrt{\frac{\beta e^3 E}{4\pi\epsilon_0}}\right)/k_{\mathrm{B}}T}$$

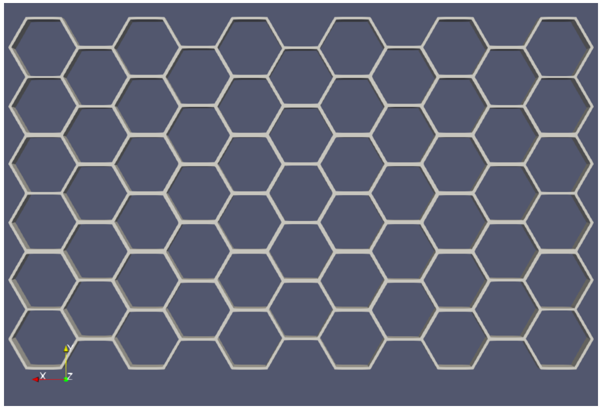


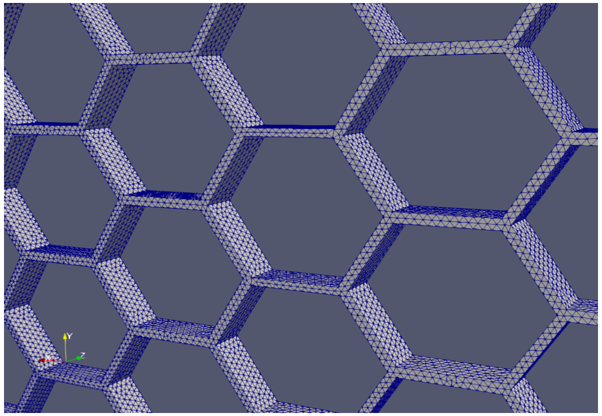


CAD interface to Warp

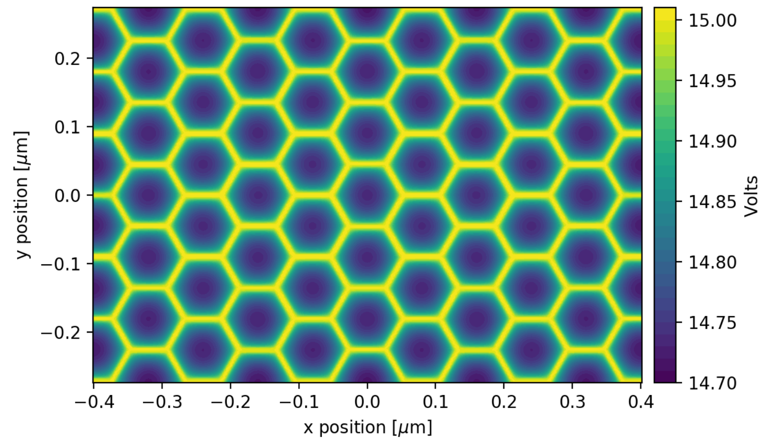
- We have modified Warp to permit standard file imports
  - *Open source package PyMesh adapted for Warp's conductor interface*
  - *Ray-triangle intersection algorithm determines representation on mesh*

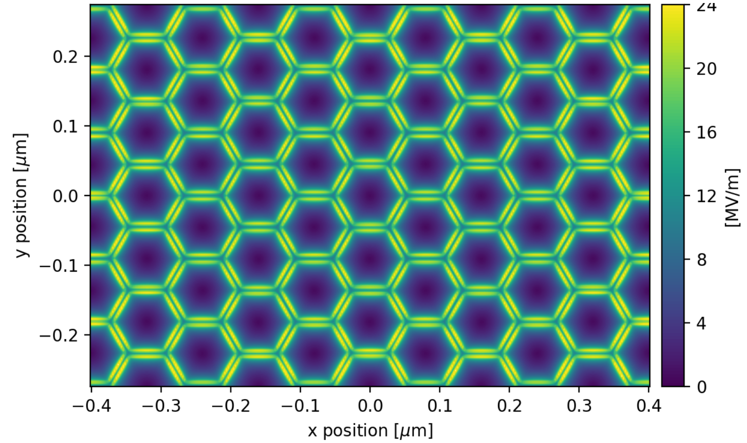
1. Design, mesh, and export externally





2. Import and install in Warp





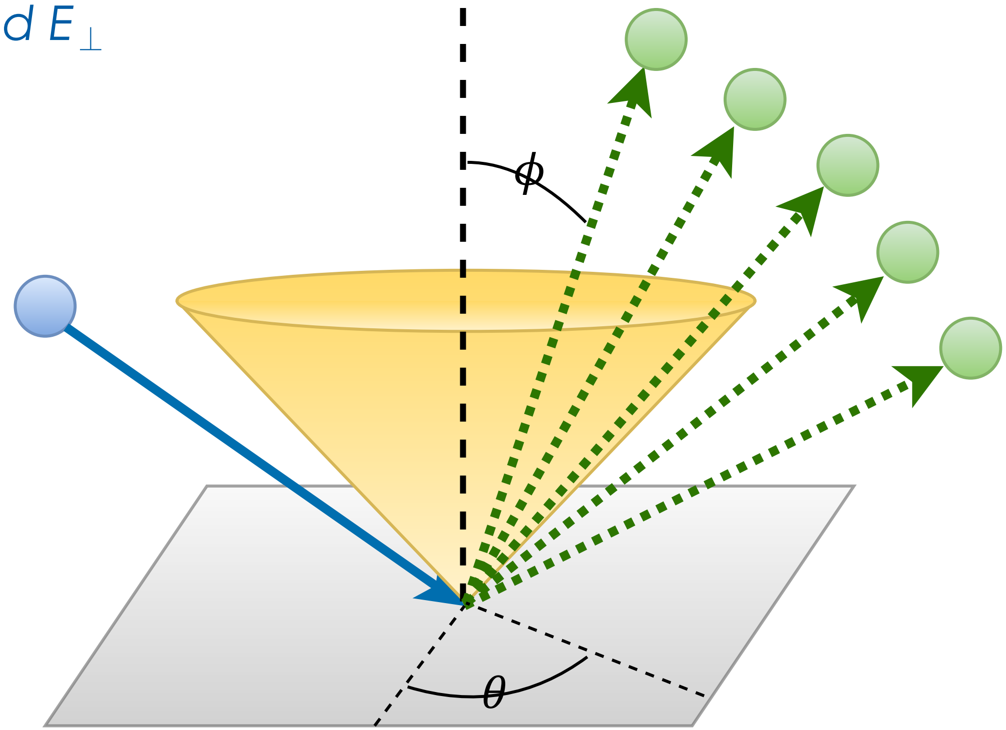
Modeling particle reflection for TECs

- Diffuse reflection has been demonstrated as an important mechanism for proper modeling*
- Two approaches:
  1. *Uniform – split between E_z and $E_\perp$*

$$v_z = \sqrt{\mathbf{v}^2 \times \mathrm{Rand}(0,1)}$$
$$v_x = \sqrt{(\mathbf{v}^2 - v_z^2) \times \mathrm{Rand}(0,1)}$$
$$v_y = \sqrt{\mathbf{v}^2 - v_z^2 - v_x^2}$$

  2. *Random angle (ϕ, θ)*

$$v_z = \mathbf{v}\cos(\phi)$$
$$v_x = \mathbf{v}\sin(\phi)\cos(\theta)$$
$$v_y = \mathbf{v}\sin(\phi)\sin(\theta)$$

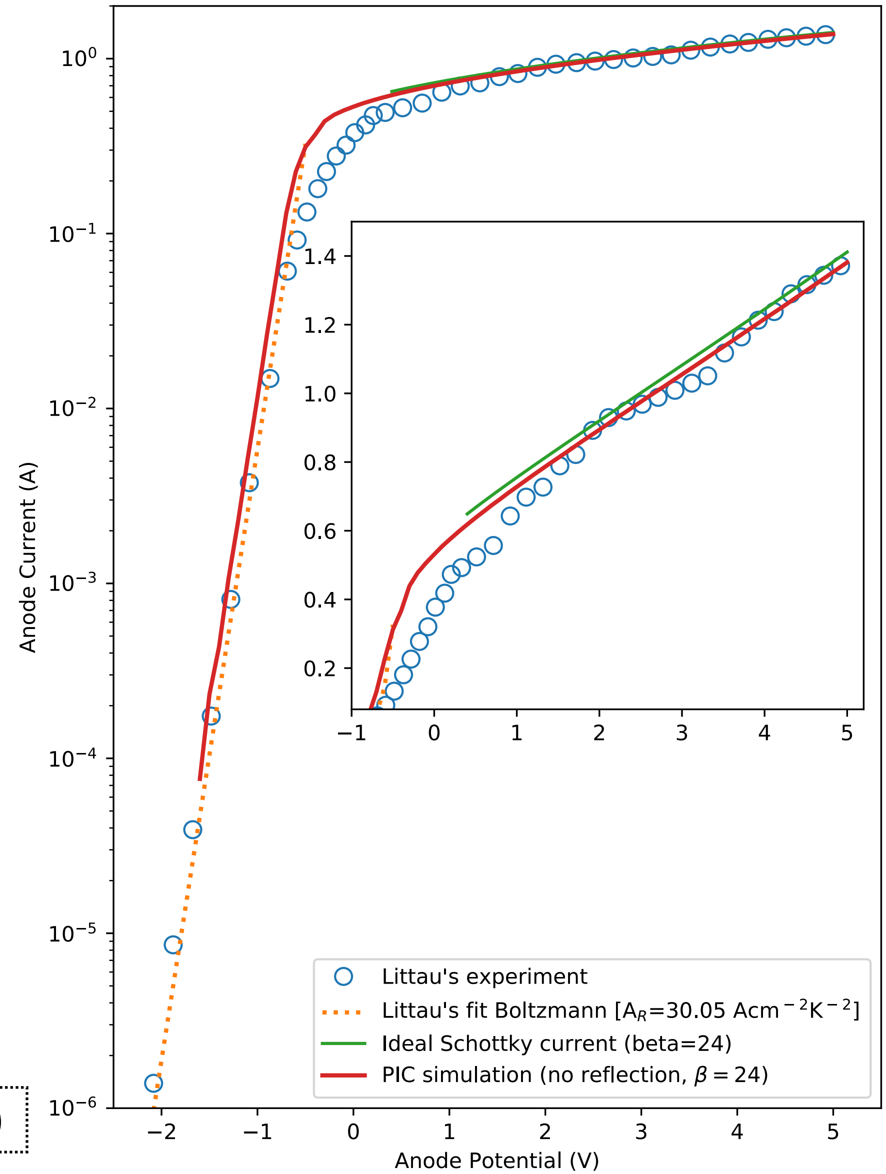


- Assumptions
  - *P_{ref} is energy independent*
  - *Classical scattering (no tunneling)*

*Balestra, Huffman, and Yang. *Electron reflection from one-dimensional potential barriers* (1978).

Particle reflection in a diode structure

- Without reflection, simulations do overestimate measured current
  - *Discrepancy is strongest in the knee, which is desired operating regime for TECs*



Littau et al. *Phys. Chem. Chem. Phys.*, **15**, 14442, (2013)

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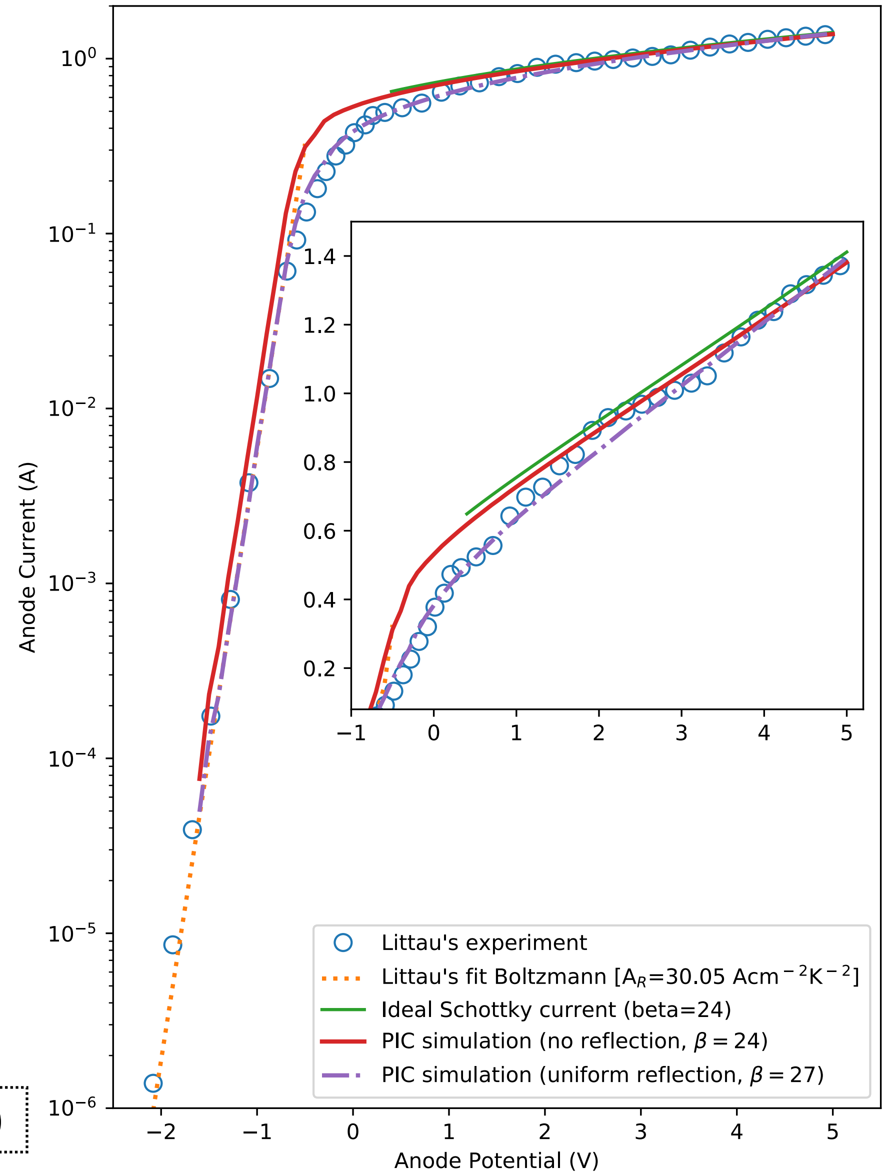
- *Particle reflection models capture reduction in current with measured geometry!*



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Particle reflection in a diode structure

- Without reflection, simulations do overestimate measured current
  - *Discrepancy is strongest in the knee, which is desired operating regime for TECs*
- *Particle reflection models capture reduction in current with measured geometry!*
- Different reflection models can produce similar results
  - *Surface characteristics are used as fitting parameters*
    - Geometric field enhancement
    - Richardson constant (λ_R)
  - *Differences in reflection probability merit further consideration*
  - *Should consider energy dependence*

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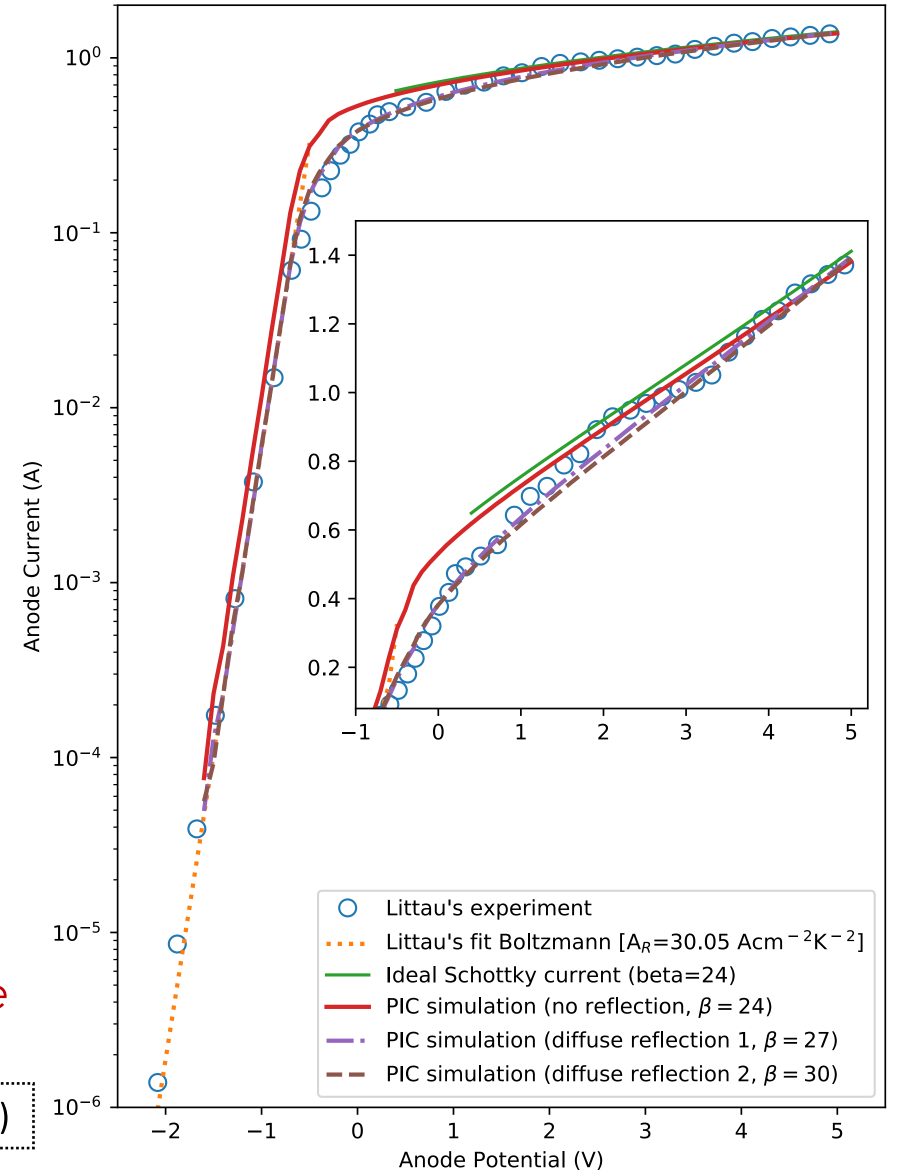
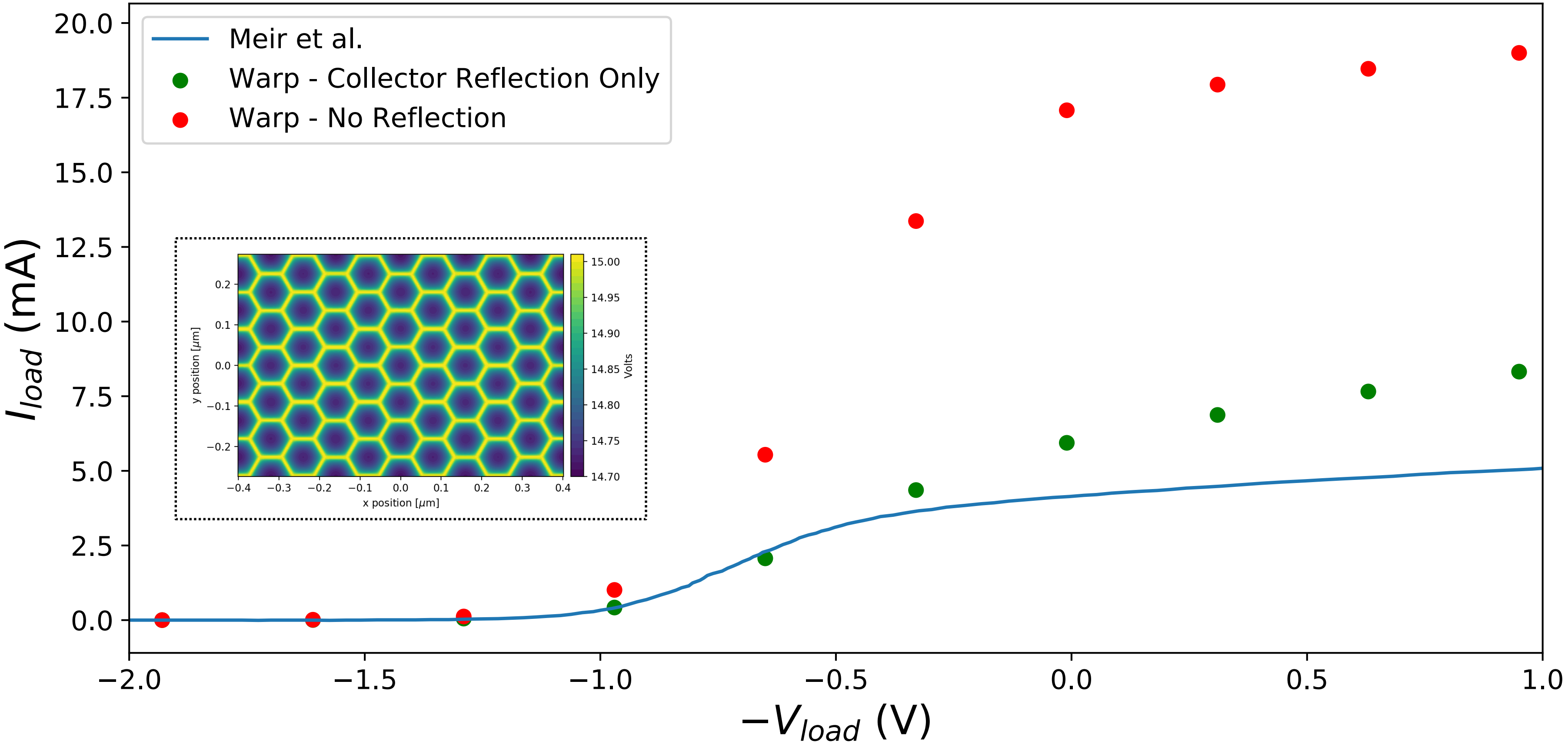


Particle reflection in a triode structure

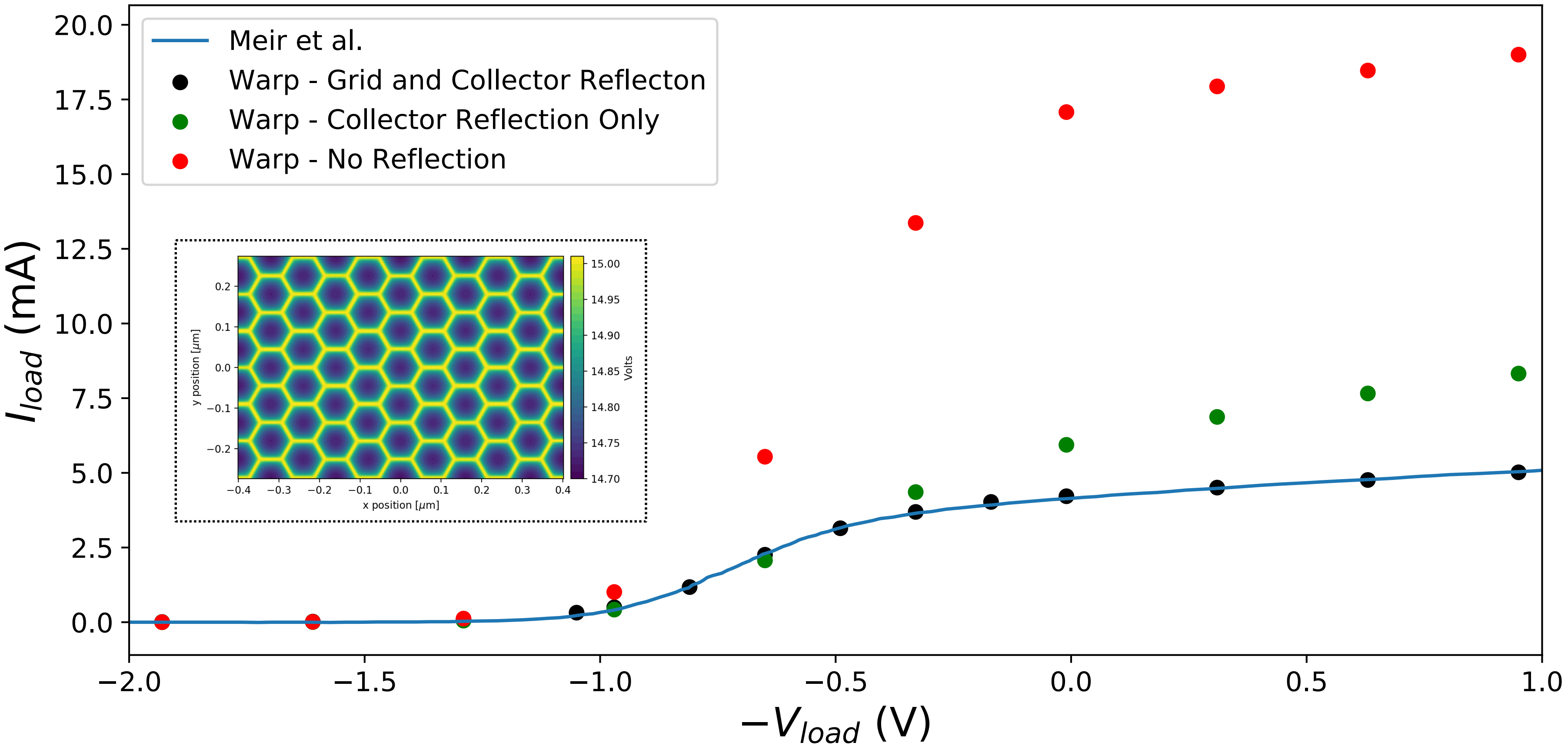
- Compare to a triode structure with honeycomb gate
- Reflection at collector is insufficient to provide agreement



Meir et al. J. Renwable Sustainable Energy, **15**, 043127, (2013)

Particle reflection in a triode structure

- Compare to a triode structure with honeycomb gate
- Reflection at gates produces good agreement!
  - *Reflection modelled probabilistically using a geometric cross-section*
  - *Full kinetic modeling is computationally demanding for STL files*



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Evaluation and Optimization of Devices

- Goal: Maximize current at collector, minimize losses

$$\eta = \frac{P_{\mathrm{load}} - P_{\mathrm{gate}}}{P_{\mathrm{ec}} + P_{\mathrm{R}} + P_{\mathrm{ew}}}$$

- Loss Channels:
  1. *Kinetic losses - excess electron kinetic energy heats surface*

$$P_{\mathrm{ec}} = J_{\mathrm{e}}\left(\phi_{\mathrm{e}} + 2k_{\mathrm{B}}T_{\mathrm{e}}\right) - \mathrm{t}J_{\mathrm{c}}\left(\phi_{\mathrm{e}} + 2k_{\mathrm{B}}T_{\mathrm{c}}\right)$$

  2. *Gate losses - electron intercepted by accelerating gate*

$$P_{\mathrm{gate}} = V_{\mathrm{gate}}\left(J_{\mathrm{gate}} + \mathrm{t}J_{\mathrm{c}}\right)$$

  3. *Radiative losses - heat lost through emissivity*

$$P_{\mathrm{R}} = \epsilon\sigma_{\mathrm{sb}}\left(T_{\mathrm{e}}^{4} - T_{\mathrm{c}}^{4}\right)$$

  4. *Resistive losses - losses in external circuit*

$$P_{ew} = 0.5\left(\frac{L}{\rho_{ew}}(T_e - T_{env})^2 - \rho_{ew}(J_{ec} - tJ_c)^2\right)$$

For more on these models, see Voesch et al. *Energy Technology*, **5**(12):2234-2243, (2017) and Cook et al. *Proc. ICAP'18,* 59-63, (2018)

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2. *Gate losses - electron intercepted by accelerating gate*

$$P_{\mathrm{gate}} = V_{\mathrm{gate}} \left(J_{\mathrm{gate}} + \mathrm{t}J_{\mathrm{c}}\right)$$

Compute these through particle statistics and tracking.

For more on these models, see Voesch et al. *Energy Technology*, **5**(12):2234-2243, (2017) and Cook et al. *Proc. ICAP'18*, 59-63, (2018)

Evaluation and Optimization of Devices

- Goal: Maximize current at collector, minimize losses

$$\eta = \frac{P_{\rm load} - P_{\rm gate}}{P_{\rm ec} + P_{\rm R} + P_{\rm ew}}$$

- Loss Channels:

Compute through semi-analytic models and iteration.

3. *Radiative losses - heat lost through emissivity*

$$P_{\rm R} = \epsilon\sigma_{\rm sb}\left(T_{\rm e}^4 - T_{\rm c}^4\right)$$

4. *Resistive losses - losses in external circuit*

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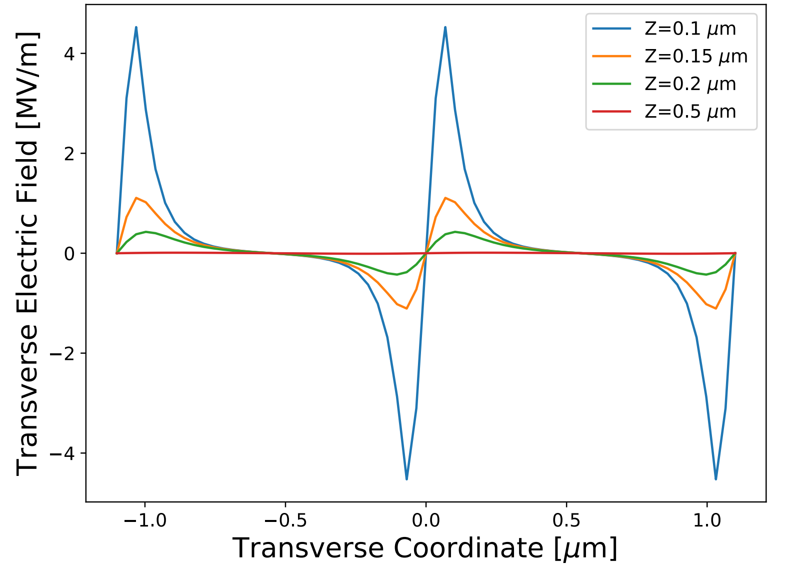
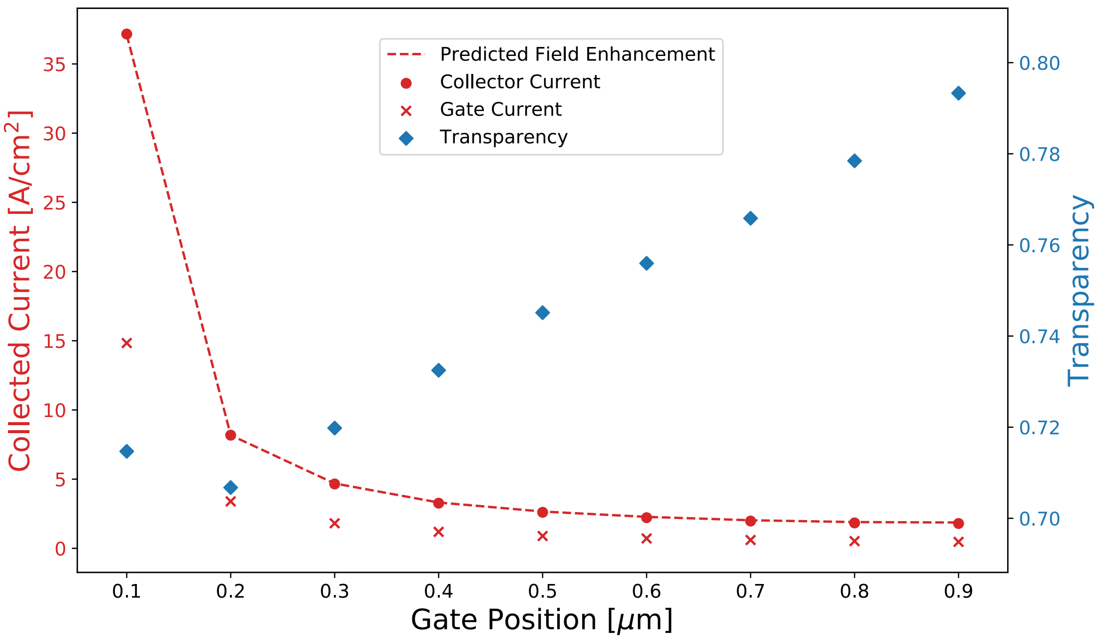
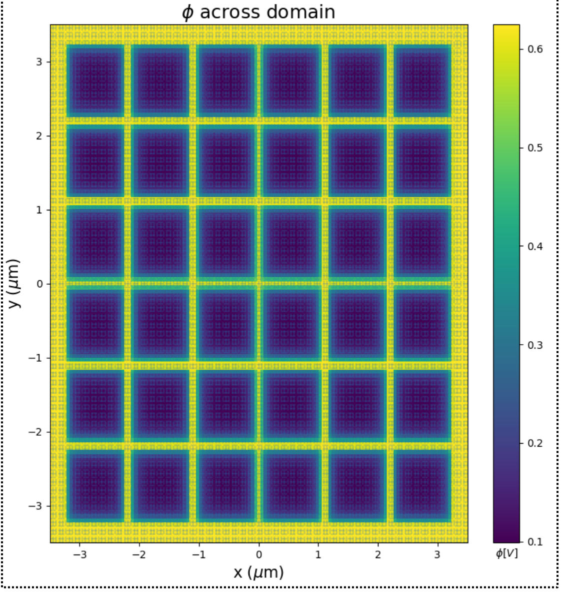
Case study in gate losses

- Gate location is critical
  - *Balance transverse dynamics with applied voltage*
  - *Gate design is constrained by practical concerns*
  - *Nanomaterials are promising*

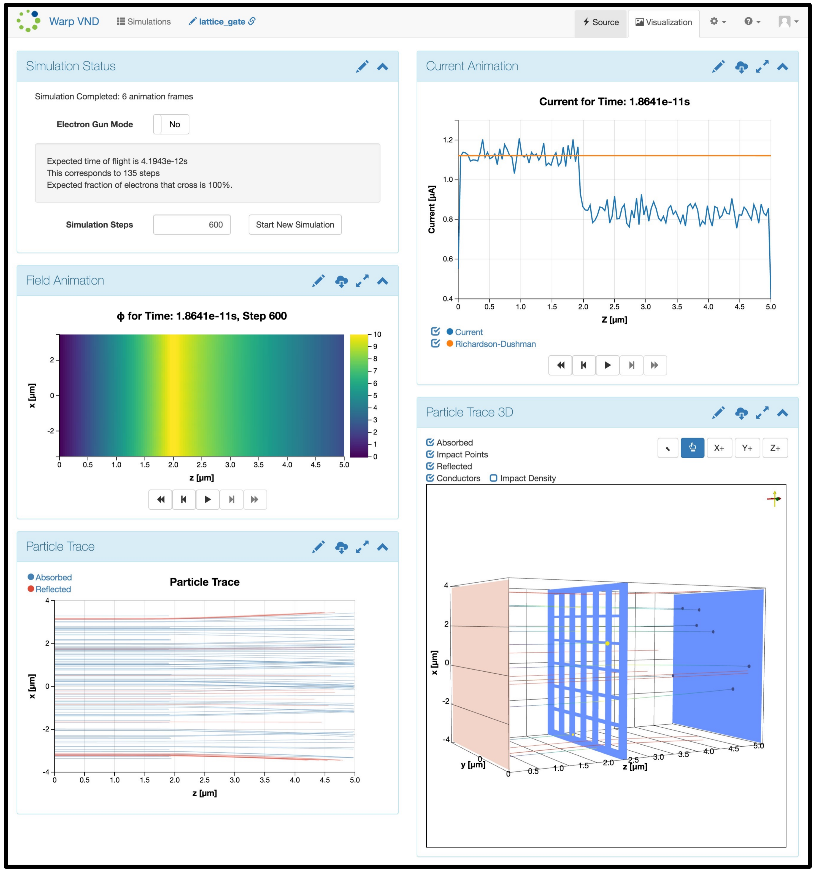


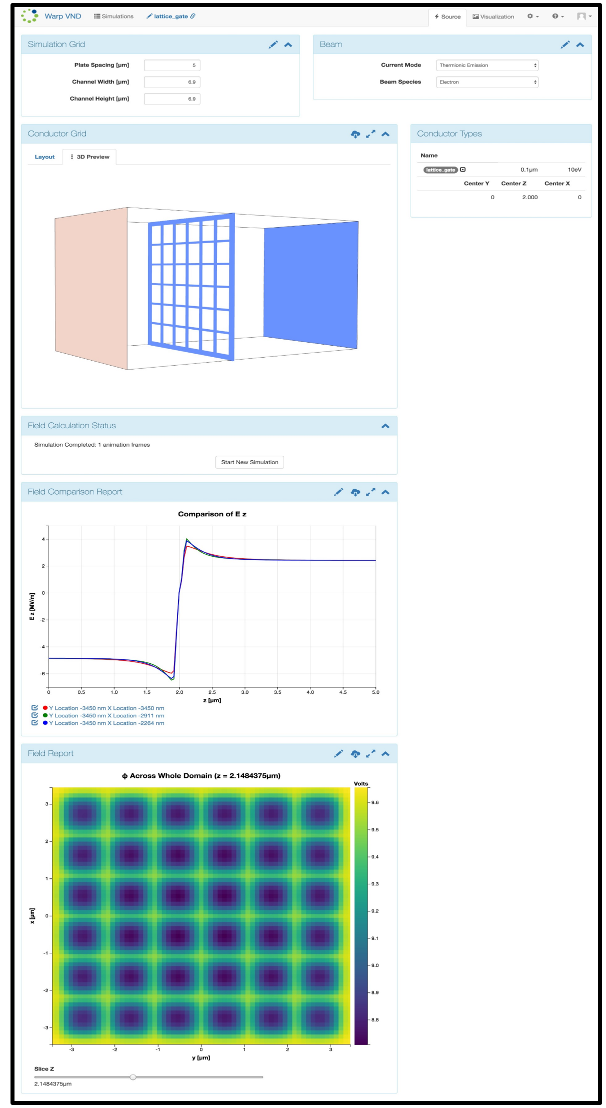




A cloud-based platform for Vacuum Nanoelectornics

- RadiaSoft's Sirepo platform hosts Warp – https://www.sirepo.com/warpvnd
- CAD interface for STL format – upload and render in Warp
- Diagnostics for fields, currents, particle loss
- Choice of solvers and geometry (2D/3D/quasi-static)
- 2D/3D visualizations via VTK.js
- Share with a link, or download Python script.





Conclusion

- We are improving the Warp code to simulate TECs with high fidelity, including the following enhancements:
  - *Internal dielectrics with 3D geometries*
  - *CAD import from standard file types*
  - *Mesh refinement and `cut-cell` capabilities*
- We've combined these tools with domain-specific physics models to capture realistic device operation:
  - *Particle reflection from collector and gate boundaries*
  - *Efficiency metrics for device operation*
  - *External circuit models for variable anode biasing*
- These tools have enabled improved benchmarks of contemporary devices against measurement
- More work is needed to improve emission characteristics, reflection, and non-classical effects